

Early Detection of ASD (Autism Spectrum Disorder) in Toddler with a Web Application

Farrel Jonathan Vickeldo

Biomedical Engineering

School of Electrical Engineering and Informatics

Bandung, Indonesia

18320008@mahasiswa.itb.ac.id

Allya Paramita Koesoema, S.T., M.T., M.Sc., Ph.D.

Biomedical Engineering

School of Electrical Engineering and Informatics

Bandung, Indonesia

allya.koesoema@itb.ac.id

dr. Marietta Shanti Prananta Sp.KFR Ped (K)

Physical Medicine and Rehabilitation

Faculty of Health, Padjadjaran University

Bandung, Indonesia

mariettashanti@gmail.com

Abstract—Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder that affects an individual's social and communication skills and their family life. Worldwide, around 168 million people are affected by ASD, including 242,541 children in Indonesia in 2022. ASD symptoms can be recognized as early as 12-24 months of age if examined by a medical professional. However, lack of parental knowledge causes delays in ASD detection, which has an impact on the increasing number of ASD cases that are detected late. Early detection of ASD can reduce the severity of long-term symptoms with children who are diagnosed early showing a decrease in social symptoms within 1-2 years. Preschool teachers have the potential to detect abnormal development in toddlers but have not been directly involved due to lack of training and screening tools. The Tiny Detector system is designed to help preschool teachers conduct early screening for ASD using the M-CHAT questionnaire. This system consists of three main components: a Flutter-based front-end service, a Node.js-based back-end service, and a PostgreSQL-based DBMS. The implementation of Tiny Detector showed an average usability score of 64.83%, efficiency of 80.77%, effectiveness of 78.21%, satisfaction of 73.69%, learnability of 26.40%, error occurrence of 25%, and cognitive load of 45.08%. This system increases the confidence and ability of preschool teachers in identifying the risk of ASD in toddlers and allows for faster early detection so that children with ASD can receive treatment as soon as possible.

Index Terms—ASD, Tiny Detector, teacher, toddler, early detection

I. BACKGROUND

Autism Spectrum Disorder (ASD) is a neurodevelopmental disorder that affects an individual's social and communication skills, as well as impacting their family life [1]. Worldwide, ASD affects around 168 million people, and in 2022, Indonesia had 242,541 children with autism, making it the country with the sixth largest population of children with autism in the world [2]. ASD symptoms can be recognized from the age of 12-24 months, but late detection, especially in toddlers, results in late treatment and an increase in the number of ASD cases [3]. Research shows that children with ASD who

are diagnosed early show a decrease in the severity of social symptoms within 1-2 years [4].

ASD screening plays a vital role in speeding up treatment in toddlers. It can raise awareness of the potential risks of ASD, so toddlers can receive the treatment they need sooner. Many toddlers don't receive a formal diagnosis until they are five years old or older, making early screening crucial [5]. Children with ASD tend to experience delays in language development and have restricted and repetitive behavior patterns, which have a negative impact on their social lives [4]. This delay in diagnosis is often due to the lack of visible early symptoms and low parental awareness of the risk of ASD. Currently, ASD screening is carried out by medical professionals based on parental concerns, but this process is time-consuming and involves many complex questions and assessments [6].

Preschool teachers, such as kindergarten and early childhood education teachers, can play a strategic role in detecting abnormal development in toddlers because they often meet and understand child development. However, preschool teachers are currently not directly involved in detecting ASD due to the lack of training and available screening tools. Research shows that the main barriers for preschool teachers to detect ASD early are the lack of knowledge and adequate screening tools [7].

Various questionnaire-based ASD screening methods have been developed, such as the Modified Checklist for Autism in Toddlers, Revised (M-CHAT-R), Childhood Autism Rating Scale, 2nd Edition (CARS2), and Developmental Behavior Checklist – Early Screen (DBC-ES) [8]. However, this method cannot yet be used by the general public due to the complexity of the assessment guide that requires a deep understanding of certain cases. Preschool teachers need socialization and basic knowledge regarding ASD, symptoms, the impact of delayed treatment, and how to use the screening method.

Research shows that there are significant differences between the development of competencies in children with ASD compared to children who are typically developing [9]. Chil-

dren with ASD have lower achievement scores in language, mathematics, and world orientation at the end of primary school. Delayed diagnosis impacts a child's social life, which can lead to anxiety and depression in adolescence.

Currently, applications that implement the questionnaire method for ASD screening are still few in several countries, such as ASDTracker from Australia and Autism Fingerprint from Oman. Applications are a good solution because currently almost everyone has a smartphone, including teachers and can be a means of improving teachers' abilities and experiences regarding ASD because they are equipped with information features. However, the existing applications do not yet have an Indonesian translation.

Therefore, the problem of ASD detection can be reduced if there is an educational application that is easy for preschool teachers to use to screen for ASD in toddlers. This final project aims to build a screening system for early detection of ASD using a questionnaire method, so that more toddlers are detected with ASD earlier, thereby reducing the burden of treatment. This system is expected to provide knowledge and awareness of the risks of ASD to preschool teachers, so that they can communicate it to the parents of toddlers and detect the risk of ASD in toddlers early. Ultimately, this system is expected to help professional medical personnel treat ASD patients earlier.

II. LITERATURE REVIEW

This section describes a literature review and comparative analysis of currently existing ASD screening systems.

A. Modified Checklist for Autism in Toddlers, Revised with Follow-Up (M-CHAT-R/F)

M-CHAT-R/F is a screening tool that has the main goal of maximizing sensitivity by detecting as many cases of ASD as possible. The results of the M-CHAT assessment will show the risk of ASD in a score. Of the 20 question items, if the assessment response shows "no" to questions other than questions 2, 5, and 12, it will increase the ASD risk score. ASD risk classification is shown in Table I [10].

TABLE I
ASD RISK CLASSIFICATION BASED ON M-CHAT-R/F

Risk	Score	Description
Low	0 – 2	If child is younger than 24 months, screen again after second birthday. No further action required unless surveillance indicates risk for ASD.
Medium	3 – 7	Administer the Follow-Up (second stage of M-CHAT-R/F) to get additional information about at-risk responses. If M-CHAT-R/F score remains at 2 or higher, the child has screened positive. Action required: refer child for diagnostic evaluation and eligibility evaluation for early intervention. If score on Follow-Up is 0-1, child has screened negative. No further action required unless surveillance indicates risk for ASD. Child should be rescreened at future well-child visits.
High	8 – 20	It is acceptable to bypass the Follow-Up and refer immediately for diagnostic evaluation and eligibility evaluation for early intervention.

B. Current ASD Screening Systems

a) *ASDTests*: The application is designed using four main questionnaire methods: Q-CHAT-10, AQ-Child-10, AQ-Adolescent-10, AQ-Adult-10. ASDTests can be used by families, caregivers, and academic teachers. This application will show the assessment results in the form of ASD status of the assessment target. A unique feature of this application is that it is available in 10 languages. In addition, this application uses machine learning (ML) in the form of a logistic regression model to show important information in screening. This application also provides basic information about ASD: symptoms, causes, and potential ASD interventions [6].

b) *Autism AI*: A proposed application that uses ML to aid in the diagnosis and support of ASD. However, since it is still in the research and development stage, its features and functions are not yet fully known. This application can be used by families, caregivers, clinicians, medical staff, teachers, and individuals with standard IQ and above. This application is used to assess toddlers and adults using the Q-CHAT-10 and AQ-10 methods. The assessment results from this application are in the form of a percentage of similarity of ASD cases. This application is based on the Android operating system. A unique feature of this application is the use of ML to analyze the behavioral data of the assessment target [6].

c) *ASDetect*: A mobile application designed to provide parents and teachers of children aged 11 to 30 months with the ability to assess the likelihood of autism identified in SACS (Social Attention and Communication Surveillance). There are 10 questions that will show the risk of severity. A unique feature of this application is the use of video visualizations containing scenarios for each question. In addition, the assessment can be continued at a later time [5].

d) *Autism Fingerprint*: An application developed in 2013 based on the Apple iOS operating system using M-CHAT and adapted into Arabic. This application is used by caregivers to assess toddlers. This application has 20 questions with the assessment results in the form of severity risk. A unique feature of this application is the use of animation and comparative images in each question by adapting Omani culture in each image. This application is available in Arabic and English. This application also provides basic information about ASD [11].

III. PROPOSED SOLUTION: TINY DETECTOR

This section describes the designed Tiny Detector system and its components and the system journey. Tiny Detector is a web-based system that aims to help preschool teachers detect ASD early in toddlers.

A. System Architecture

The proposed system has 2 users based on functional needs, namely assessors (preschool teachers) and admins, as shown in Fig . 1.

The functional needs of the Tiny Detector system are accommodated by the main components contained in the Tiny Detector system, namely the front-end using Flutter, the

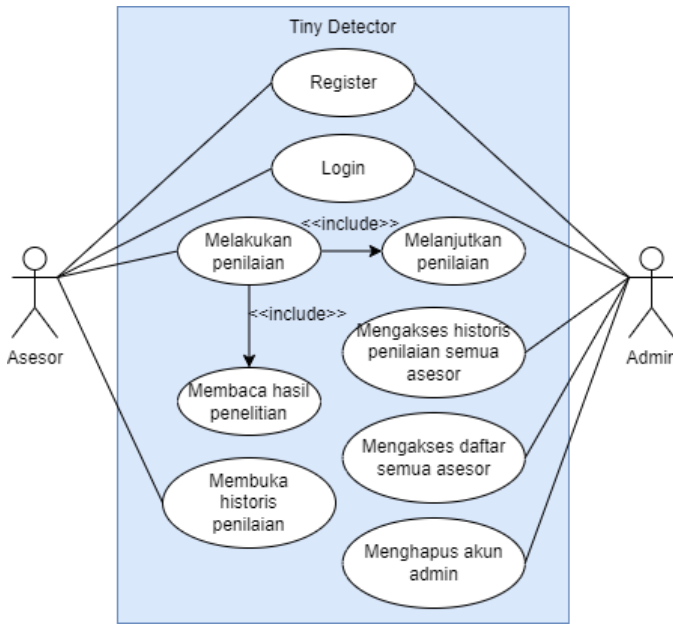


Fig. 1. Use case diagram of Tiny Detector system.

back-end using Node.js which implements integration with other components using the REST API, DBMS (Database Management System) using PostgreSQL, and VM (Virtual Machine) using Hostinger. The system architecture component diagram can be seen in Fig . 2

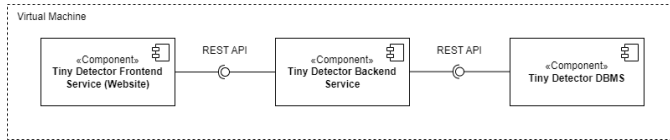


Fig. 2. Component diagram of Tiny Detector architecture.

B. User Interaction

Overall, the system journey for the assessor as a user can be divided into 3 phases: before assessment, assessment, and after assessment. In the section before assessment, after the assessor enters the main page and ensures that the test subject is under 5 years old, the assessor selects the 'Assessment Test' feature. The page will move to the toddler form and will be filled in by the assessor. The data will be sent to the server to be verified whether it is already in the database or not. After validation, the assessor will answer 20 questions with a "Yes" or "No" response. If all questions have been answered, the system will calculate the total score obtained, then save all question responses and the total score obtained. After being saved, the assessment results will be saved and displayed on the system. If there are questions that have not been answered, the system will save the progress of the question responses and display a list of assessments that have not been completed. The system journey flow can be seen in Fig 5.

IV. IMPLEMENTATION

The implementation is in the form of a website display that is built with a total of 11 pages in this system: 8 pages for assessor users (Fig 3) and 4 pages for admin users (Fig 4). Back-end integrates Front-end and DBMS so that the system can store data in real-time. The system can be accessed at www.tinydetector.com.

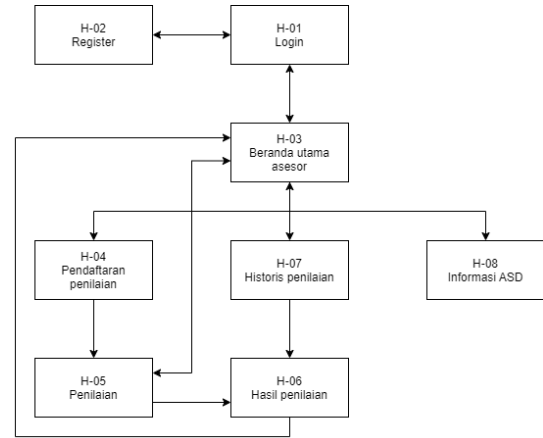


Fig. 3. Page navigation diagram for assessor users of Tiny Detector.

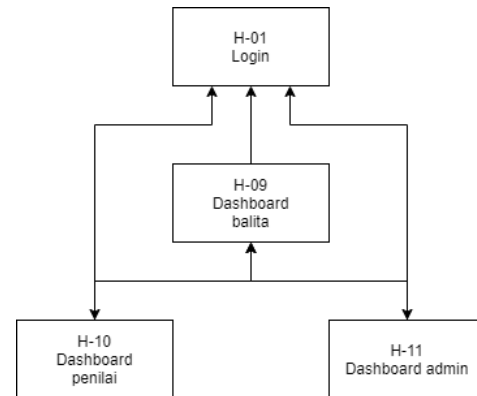


Fig. 4. Page navigation diagram for admin users of Tiny Detector.

V. TESTING

This section contains tests of self-efficacy and usability of the Tiny Detector system.

A. Method

There are 19 preschool teachers from Bandung City who are the test samples. The test begins with an ASD matriculation session to equalize the knowledge of each sample. After that, session 1 testing will begin to test self-efficacy. The results of session 1 testing are the level of sample confidence based on a particular method. The next test is continued in sessions 2 and 3 which test the usability of the system using the PACMAD (People At the Center of Mobile Application Development) method. The results of session 2 and 3 testing are a complete assessment of the usability of the system that has been

designed and improvements that can be made to improve the usability of the system. The test results obtained will be analyzed for improvement and further research development. The test flow diagram can be seen in Fig 6.

B. Results

a) *Self-efficacy*: Based on the results of session 1, the average results of self-confidence in each video and each group can be seen in Table II. Based on the results of the 1st session test, the average self-confidence in each group was 4.75 in group A and 4.1 in group B. This means that group A, namely the group that assessed the risk of ASD indications in toddlers in the video with intuition, had higher self-confidence compared to group B, namely the group that assessed the risk of ASD indications in toddlers with M-CHAT. This may be due to the previous matriculation session that increased teacher knowledge and the difficulty of using M-CHAT without prior socialization for laypeople.

TABLE II
TESTING RESULT OF SESSION 1

Video	Group A	Group B	Average of each group
1	5,1	4,9	5
2	3,7	3,3	3,5
3	5,1	3,8	4,45
4	5,1	4,4	4,75
Average of each group	4,75	4,1	4,43

b) *Efficiency*: Efficiency is measured by calculating the time it takes users to successfully complete a task compared to the total time it takes to complete all tasks. Efficiency in session 3 showed an increase in each age group. Based on the test results, the average efficiency in each session was 72.93% in session 2 and 88.61% in session 3. The results of the efficiency test can be seen in Table III. The number of tasks that could not be completed also decreased in session 3. Based on the test results, the total number of tasks that were not successfully completed in each session was 40 tasks in session 2 and 18 tasks in session 3 with the total number of tasks that were not successfully completed in each sample being 58 tasks. The results of the efficiency test can be seen in Table IV.

TABLE III
RESULTS OF USABILITY TESTING OF EFFICIENCY ATTRIBUTES IN 2 SESSIONS

Age group	Session 2 (%)	Session 3 (%)	Average for each age group (%)
Under 25 years	81,04	100	90,52
25 – 35 years	85,38	85,58	85,48
36 – 45 years	77,08	89,86	83,47
Above 45 years	48,21	78,99	63,60
Average per session (%)	72,93	88,61	80,77

TABLE IV
NUMBER OF FAILED ASSIGNMENTS IN 2 SESSIONS

Age group	Session 2	Session 3	Total for each age group
Under 25 years	4	0	4
25 – 35 years	3	3	6
36 – 45 years	11	4	15
Diatas 45 years	22	11	33
Total per session	40	18	58

c) *Effectiveness*: Effectiveness is measured by calculating the number of tasks successfully performed by users compared to the total number of tasks. Effectiveness in session 3 showed an increase in each age group, except for the 25-35 age group. This can be caused by human error and difficulty in improving performance. Based on the test results, the average effectiveness in each session was 71.49% in session 2 and 84.93% in session 3. The results of the effectiveness test can be seen in Table V.

TABLE V
RESULTS OF USABILITY TESTING OF EFFECTIVENESS ATTRIBUTES IN 2 SESSIONS

Age group	Session 2 (%)	Session 3 (%)	Average for each age group (%)
Under 25 years	85,71	100	92,86
25 – 35 years	89,29	87,5	88,40
36 – 45 years	73,81	88,89	81,35
Above 45 years	37,14	63,33	50,24
Average per session (%)	71,49	84,93	78,21

d) *Satisfaction*: Satisfaction was measured using the SUS (System Usability Scale) method by answering 10 questions with Likert 1-5. Based on the test results, the average satisfaction reached a score of 73.69 with a 'good' rating. The results of the satisfaction test can be seen in Table VI.

TABLE VI
RESULTS OF USABILITY TESTING OF EFFECTIVENESS ATTRIBUTES IN 2 SESSIONS

Age group	Average for each age group	Rank	Description
Under 25 years	75	B	Good
25 – 35 years	77,5	B	Good
36 – 45 years	76,25	B	Good
Above 45 years	66	D	Not good
Average satisfaction	73,69	B	Good

e) *Learnability*: Learnability was measured using the effectiveness value in both sessions. Learnability showed an increase in each group, except for the 25-35 age group due to its positive correlation with effectiveness. Based on the test results, the average learning ability increased by 26.4%, as in Table VII.

f) *Error occurrence*: Error occurrence is measured by calculating the total errors during the task. The user's screen

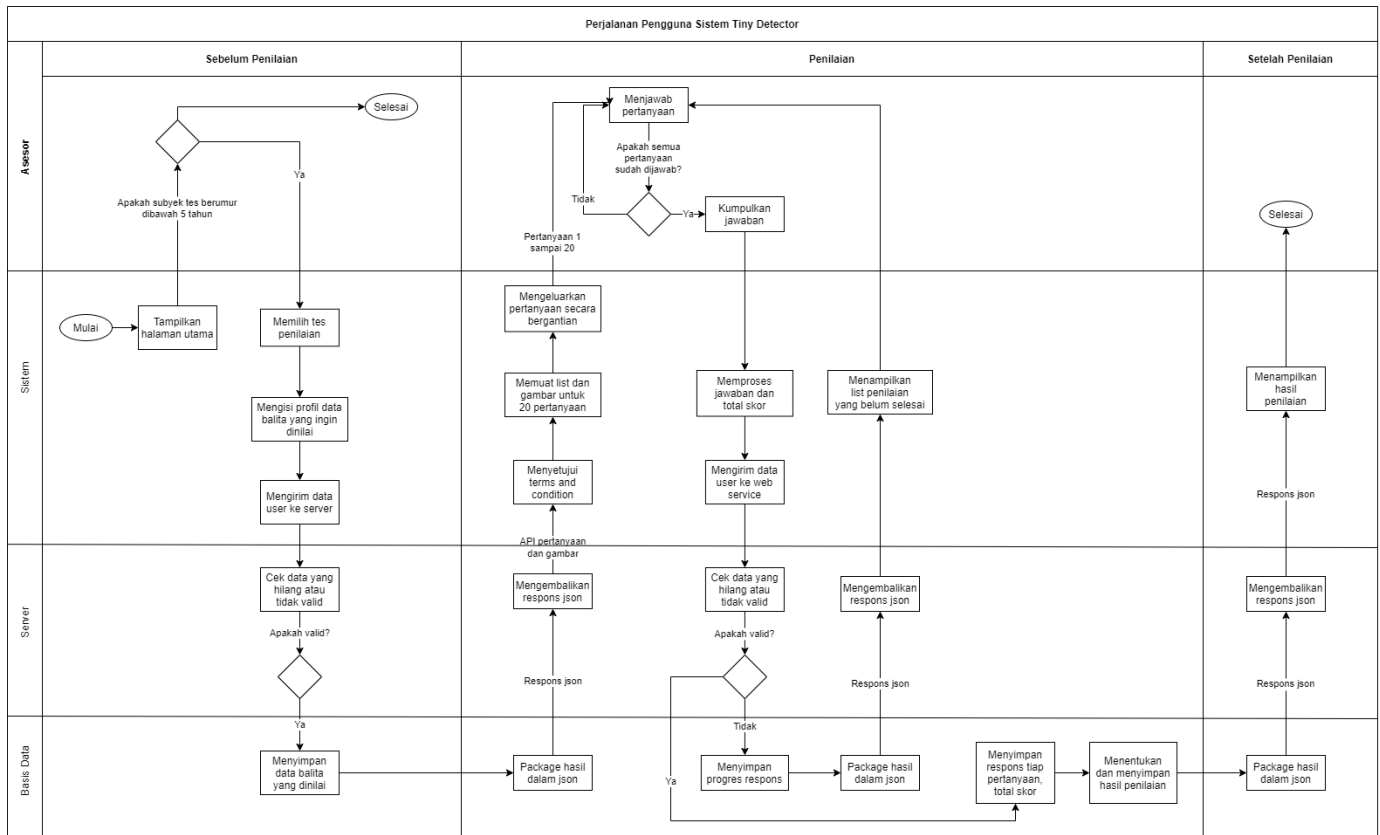


Fig. 5. The activity diagram of Tiny Detector.

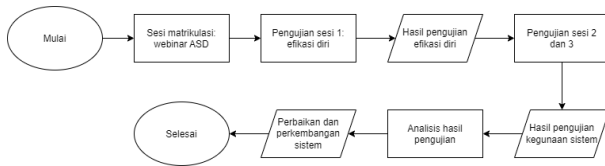


Fig. 6. Testing flowchart of Tiny Detector.

TABLE VII
RESULTS OF USABILITY TESTING OF LEARNABILITY ATTRIBUTES IN 2 SESSIONS

Age group	Effectiveness of session 2 (%)	Effectiveness of session 3 (%)	Learnability (%)
Under 25 years	85,71	100	16,67
25 – 35 years	89,29	87,5	-2
36 – 45 years	73,81	88,89	20,43
Above 45 years	37,14	63,33	70,52
Average (%)	71,49	84,93	26,4

will be recorded and the recording results will be calculated to calculate critical and non-critical errors in each session, as in Table VIII. Based on the test results, the number of errors occurred was 59 errors consisting of 26 critical errors and 20 non-critical errors in session 2 and 10 critical errors and 3 non-critical errors in session 3, as in Fig 7.

TABLE VIII
CLASSIFICATION OF ERRORS

Critical error	Noncritical error
Server time-out too often	There are buttons that are not well defined
Interface design, writing, colors, and panels make it difficult for users	There is a delay in running the system
API call not working properly	Name and email on sidebar do not match personal data
Navigation is quite difficult	Many terms and illustrations are difficult to understand

g) *Load cognitive*: Cognitive load was measured using the NASA-TLX (National Aeronautics and Space Administration Task Load Index) method. Based on the test results, the average cognitive load reached a value of 45.08 with a 'somewhat high' category, as in Table IX.

TABLE IX
RESULTS OF THE COGNITIVE LOAD ATTRIBUTE USABILITY TEST

Age group	Average value for each age group	Cognitive load classes
Under 25 years	37,62	Somewhat high
25 – 35 years	44,76	Somewhat high
36 – 45 years	43,65	Somewhat high
Above 45 years	54,29	High
Average	45,08	Somewhat high

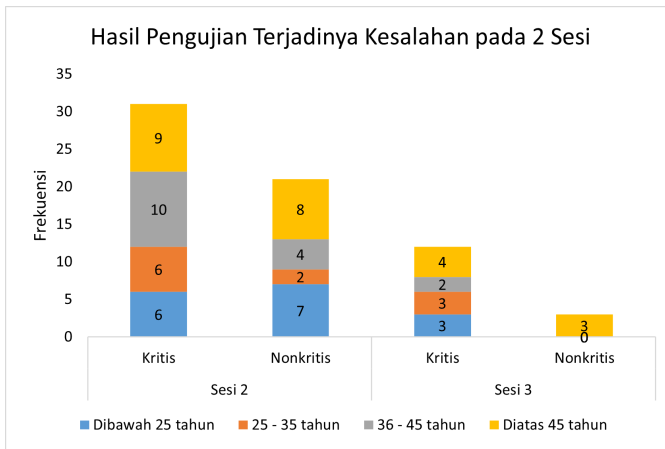


Fig. 7. Results of the attribute usability test showed errors occurring in 2 sessions.

h) Overall usability testing: Fig 8 includes data on all usability attributes of the PACMAD model along with the overall usability of the system.

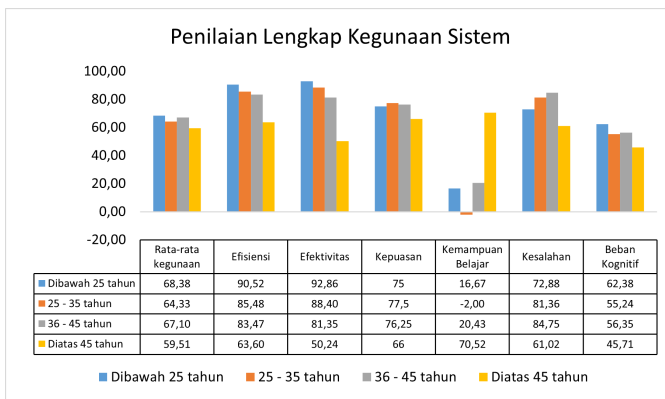


Fig. 8. Results of usability testing of error occurrence attributes in 2 sessions

VI. CONCLUSIONS AND RECOMMENDATIONS

a) Conclusions: The implementation of early screening methods for ASD by preschool teachers requires increased knowledge and experience. The use of M-CHAT has been proven effective in increasing teacher confidence and self-efficacy, with a self-efficacy score of 4.1 out of 6. The Tiny Detector system, which uses the medically recognized M-CHAT questionnaire, is designed to detect early ASD risk in toddlers, allowing for earlier intervention. The development of the system through the stages of requirement gathering and design according to the ISO/TS 82304-2 standard ensures ease of use. The test results show that this system meets the ease of use standards with an average usability score of 64.83%, efficiency of 80.77%, effectiveness of 78.21%, and satisfaction of 73.69% (ranked 'good'). Learning ability was assessed at 26.40%, error occurrence 25%, and cognitive load 45.08 (ranked 'rather high'). This system has been proven easy

to use by preschool teachers for early screening for ASD, in accordance with the methods commonly used in Indonesia.

b) Recommendations: Focus development on improving existing systems before building new ones, including improving security through encryption, authentication, and improving the decode token process. Add key features as needed by users, such as information about nearby hospitals, and improve the interface to meet the needs of a diverse user base. Additionally, improve navigation by implementing a single page application and better cookie management. Ensure continued funding for the hosting and deployment of the system by choosing an efficient and quality hosting service..

REFERENCES

- [1] H. Hodges, C. Fealko, and N. Soares, "Autism spectrum disorder: definition, epidemiology, causes, and clinical evaluation," *Translational Pediatrics*, vol. 9.
- [2] Autism rates by country 2023 - wisevoter, <https://wisevoter.com/country-rankings/autism-rates-by-country/>.
- [3] American Psychiatric Association 2013, *Diagnostic and statistical manual of mental disorders : DSM-5*, American Psychiatric Association, Washington DC, 5th ed., 2013.
- [4] N. Gabbay-Dizdar, M. Ilan, G. Meiri, M. Faroy, A. Michaelovski, H. Flusser, I. Menashe, J. Koller, D. A. Zachor, I. Dinstein, "Early diagnosis of autism in the community is associated with marked improvement in social symptoms within 1–2 years," *Sage Journals*, vol. 26.
- [5] J. Barbaro and M. Yaari, "Study protocol for an evaluation of ASDETECT - a mobile application for the early detection of autism," *BMC Pediatrics*, vol. 20.
- [6] S. R. Shahamiri and F. Thabtah, "Autism AI: A new autism screening system based on Artificial Intelligence," *Cognitive Computation*, vol. 12.
- [7] S. Tareh, N. A. Ahmad, S. Roslan, A. M. Ma'rof, S. Zaid, "Pre-School Teachers' Knowledge, Belief, Identification Skills, and Self-Efficacy in Identifying Autism Spectrum Disorder (ASD): A Conceptual Framework to Identify Children with ASD," *Brain Sciences*, vol. 10.
- [8] F. F. Gharamaleki, B. Bahrani, J. Masumi, "Autism screening tests: A narrative review," *Journal of Public Health Research*, vol. 11.
- [9] A. Golos, H. Ben-Zur, S. I. Chapani, "Participation in preschool activities of children with autistic spectrum disorder and comparison to typically developing children," *Research in Developmental Disabilities*, vol. 127.
- [10] "ChatTM - autism screening," M, <https://www.mchatscreen.com/>.
- [11] T. J. Klein, T. Al-Ghasani, M. Al-Ghasani, A. Akbar, E. Tang, Y. Al-Farsi, "A mobile application to screen for autism in Arabic-speaking communities in Oman," *The Lancet Global Health*, vol. 1.